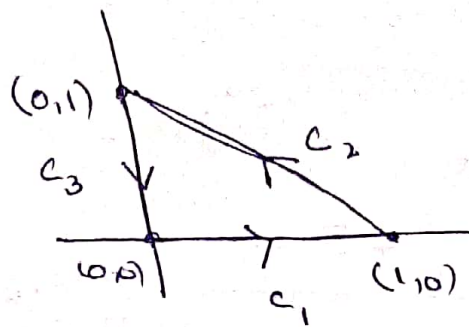


Ex

①

$$\int_C \operatorname{Im}(z^2) dz$$

①



Sol

Q1 Geometric representation

$$0 = 0 + 0i = (0,0)$$

$$1 = 1 + 0i = (1,0)$$

$$i = 0 + i = (0,1)$$

$$\int_C \operatorname{Im}(z^2) dz = \int_{C_1} \operatorname{Im}(z^2) dz + \int_{C_2} \operatorname{Im}(z^2) dz + \int_{C_3} \operatorname{Im}(z^2) dz$$

②

Q2

$$f(z) = \operatorname{Im}(x^2 - y^2 + 2ixy)$$

$$\boxed{f(z) = 2xy} \quad \text{--- (3)}$$

As $v(x,y) = 0$

so, we say say

$f(z)$ is not analytic

Q3

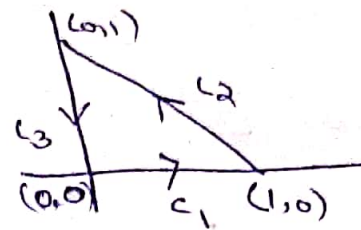
Path dependent

$$\int_C f(z) dz = \int_{t=a}^{t=b} [f(z(t)) \cdot \dot{z}(t)] dt \quad \text{--- (4)}$$

To solve (4)

$$C_1: \boxed{z(t) = x(t) + iy(t)} \quad (4)$$

(i) $x(t) = t$
 $y(t) = 0$
 $0 \leq t \leq 1$



$$\Rightarrow \boxed{z(t) = t \quad 0 \leq t \leq 1} \quad (5)$$

(ii) $\{f(z) \rightarrow f(z(t)), x \rightarrow x(t), y \rightarrow y(t)\}$

$$f(z) = f(z(t)) = 2x(t)y(t) \quad (6)$$

from *

$$\boxed{f(z(t)) = 0} \quad (7)$$

(iii) $\boxed{\dot{z}(t) = 1} \quad (8)$

$$\Rightarrow \int_{C_1} f(z) dz = \int_0^1 (2x(t)y(t)) \cdot (1) dt = 0 \quad (9)$$

C_3 : method #01

(i) $x(t) = 0, y(t) = (1-t)$

$$z(t) = 0 + (1-t)i$$

(ii) $f(z(t)) = 2x(t)y(t) = 0$

Since $f(z(t)) = 0$ we

can say that

$$\int_{C_3} f(z) dz = \int_0^1 (2x(t)y(t)) \cdot \dot{z}(t) dt = 0 \quad (10)$$

(i) $x(t) = 0, y(t) = t$
 t will be from 1 to 0

$$\int_{C_3} f(z) dz = \int_1^0 (2x(t)y(t)) \dot{z}(t) dt = 0 \quad (11)$$

(3)

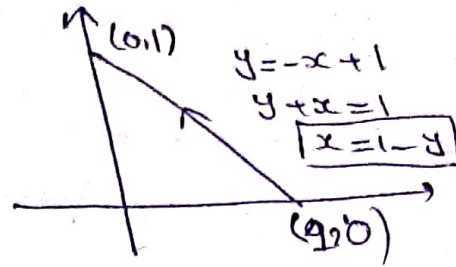
 $C_3:$

$$(i) \quad z(t) = x(t) + i y(t)$$

$$x(t) = 1-t, \quad y(t) = t$$

$\downarrow$ (1,0) $\downarrow$ (0,1)

$$0 \leq t \leq 1$$



$$(1,0) \rightarrow (0,1)$$

$$\boxed{z(t) = (1-t) + i t \quad 0 \leq t \leq 1} \quad (12)$$

$$(ii) \quad \text{from } (**) \text{ \& } (***)$$

$$f(z(t)) = 2x(t)y(t) = 2(1-t)t \quad (13)$$

$$(iii) \quad \dot{z}(t) = -1 + i \quad (14)$$

$$(iv) \quad \int_{C_2} f(z) dz = \int_0^1 (2(1-t)t \cdot (-1+i)) dt$$

$$= 2(-1+i) \left[\frac{t^2}{2} - \frac{t^3}{3} \right]_0^1$$

$$= \frac{2(-1+i)}{3} = \frac{-1+i}{3} \quad (14)$$

put (9), (10) \& (14) in (2)

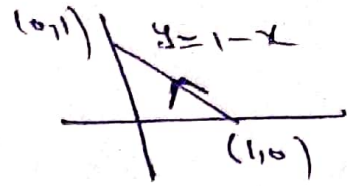
$$\int_C \operatorname{Im}(z^2) dz = 0 + 0 + \frac{-1+i}{3} = \frac{-1+i}{3}$$

(4)

(OR)

$$z(t) = t + (1-t)i$$

t is from 1 to 0.



$$t=0 \Rightarrow z=0+i=(0,1)$$

$$t=1 \Rightarrow z=1+0i=(1,0)$$

$$\Rightarrow \int_C \operatorname{Im}(z^2) dz = 2 \int_1^0 \left((t)(1-t) \right) (1-i) dt$$

$$= 2(1-i) \left[\frac{t^2}{2} - \frac{t^3}{3} \right]_1^0 = -\frac{1}{2} + \frac{1}{3} = \frac{-3+2}{6} = -\frac{1}{6}$$

$$= -\frac{2(1-i)}{6} = \frac{-1+i}{3}$$